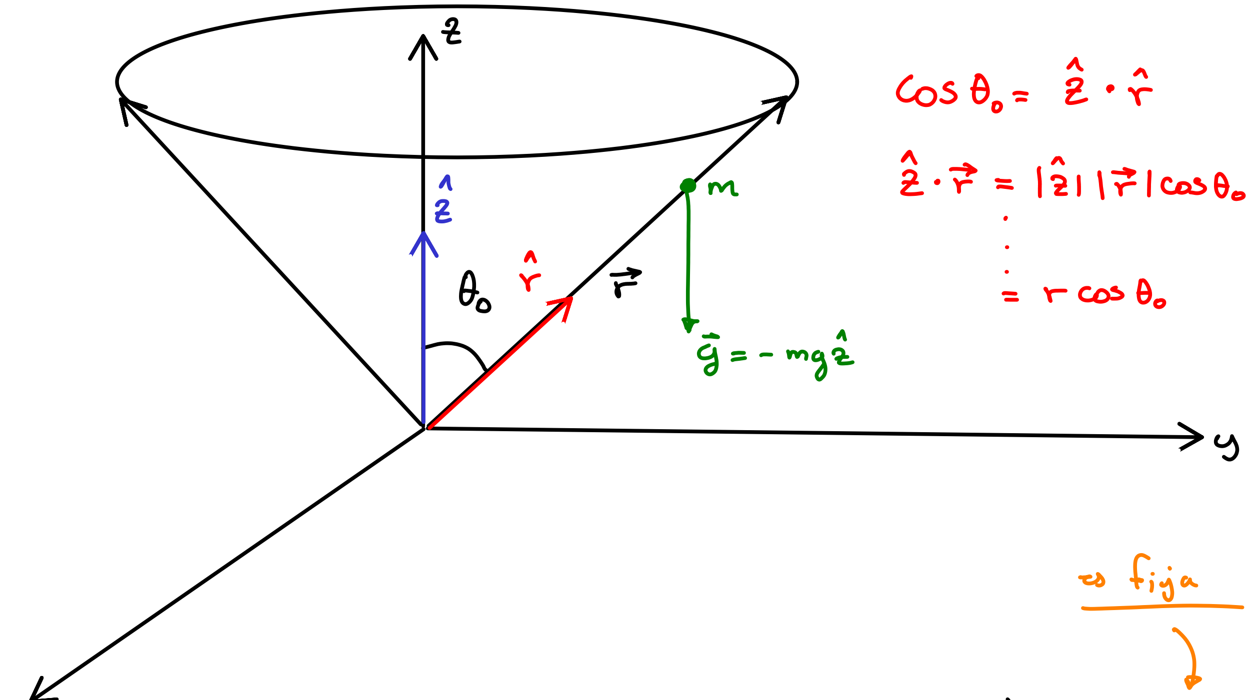
Cono




$$\cos\theta_0 = \hat{z}\cdot\hat{r}$$

$$\hat{z}\cdot\vec{r} = |\hat{z}||\vec{r}|\cos\theta_0$$

$$\vdots$$

$$= r\cos\theta_0$$

es fija

$$(x,y,z) \to (r,\theta_0,\phi) \to (r,0,\phi)$$

Tenemos ...

vector de posición (coordenadas esféricas)

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z} \quad ; \quad \left.\begin{aligned} x &= r\,\text{sen}\,\theta\cos\phi \\ y &= r\,\text{sen}\,\theta\,\text{sen}\,\phi \\ z &= r\cos\theta \end{aligned}\right\} \quad \theta = \theta_0$$

fijo es constante

Tenemos

$$\vec{r} = \underbrace{r\,\text{sen}\,\theta_0\cos\phi}_{x}\,\hat{x} + \underbrace{r\,\text{sen}\,\theta_0\,\text{sen}\,\phi}_{y}\,\hat{y} + \underbrace{r\cos\theta_0}_{z}\,\hat{z}$$

$$\dot{\theta}_0 = 0 \Rightarrow \underline{\theta_0 = \text{cte}}$$

Energía cinética...

Tenemos...
$$T = \frac{1}{2}m\left(|\dot{\vec{r}}|^2\right) = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}_0^2 + r^2\text{sen}^2\theta_0\,\dot{\phi}^2\right)$$

$$\vdots$$

$$= \frac{1}{2}m\left(\dot{r}^2 + r^2\text{sen}^2\theta_0\,\dot{\phi}^2\right)$$

Energía potencial

$$U = mgz \to mg(r\cos\theta_0)$$

Asi tenemos por $L(q_i, \dot{q}_i, t) = T - U$

Tenemos...

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2 \operatorname{sen}^2\theta_0 \dot{\phi}^2\right) - mgr\cos\theta_0$$

$$L = L(r, \phi, \dot{r}, \dot{\phi}, t)$$

Euler - Lagrange:

Tenemos $\}$ $q_1 = r$ ; $q_2 = \phi$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0 \Rightarrow \frac{d}{dt}\left(m\dot{r}\right) - \left[mr\operatorname{sen}^2\theta_0\dot{\phi}^2 - mg\cos\theta_0\right] = 0$$

$$\vdots$$

$$\Rightarrow m\ddot{r} - mr\operatorname{sen}^2\theta_0\dot{\phi}^2 + mg\cos\theta_0 = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) - \frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{d}{dt}\left(\overbrace{mr^2\operatorname{sen}^2\theta_0\dot{\phi}}^{\text{momento angular alrededor del eje del cono}}\right) = 0 \quad \left.\right\} \; mr^2\operatorname{sen}^2\theta_0\dot{\phi} = Cte$$

$\phi =$ cíclica

$$\hookrightarrow m\operatorname{sen}^2\theta_0\left[\dot{\phi}\frac{d}{dt}(r^2) + r^2\frac{d}{dt}(\dot{\phi})\right] = 0$$

$$\hookrightarrow m\operatorname{sen}^2\theta_0\left[2\dot{\phi}r\dot{r} + r^2\ddot{\phi}\right] = 0$$

Dividimos todo por $m\operatorname{sen}^2\theta_0$, tenemos

$$2r\dot{r}\dot{\phi} + r^2\ddot{\phi} = 0$$

Las dos ecuaciones de movimiento son:

$$m\ddot{r} - mr\operatorname{sen}^2\theta_0\dot{\phi}^2 + mg\cos\theta_0 = 0 \qquad (1)$$
$$2r\dot{r}\dot{\phi} + r^2\ddot{\phi} = 0 \qquad (2)$$

Hallamos la Hamiltoniana mediante la transformada de Legendre dada por

$$g(p) = F(x(p), p) = p\,x(p) - f(x(p))$$

Para el Hamiltoniano $\rightarrow$ $H = H(\dot{q}_i, p_i, t)$

Tenemos...

$$H(\dot{q}_i, \dot{p}_i, t) = \sum_{i=1}^{n} \underbrace{\frac{\partial L}{\partial \dot{q}_i}}_{p_i} \dot{q}_i - L(q_i, \dot{q}_i, t)$$

Calculamos los momentos generalizados para $\dot{q}_1 = \dot{r}$ y $\dot{q}_2 = \dot{\phi}$

$$\frac{\partial L}{\partial \dot{r}} = m\dot{r} = p_r \Rightarrow \dot{r} = \frac{p_r}{m}$$

$$\frac{\partial L}{\partial \dot{\phi}} = mr^2 \operatorname{sen}^2\theta_0 \dot{\phi} = p_\phi \Rightarrow \dot{\phi} = \frac{p_\phi}{mr^2\operatorname{sen}^2\theta_0}$$

Sustituyendo tenemos (Hamiltoniano no depende de las velocidades)

$$H(r, \phi, p_r, p_\phi, t) = m\dot{r}\dot{r} + mr^2\operatorname{sen}^2\theta_0\dot{\phi}\cdot\dot{\phi} - \left[\frac{1}{2}m\left(\dot{r}^2 + r^2\operatorname{sen}^2\theta_0\dot{\phi}^2\right) - mgr\cos\theta_0\right]$$

$$\dot{r} = \frac{p_r}{m}$$

$$\dot{\phi} = \frac{p_\phi}{mr^2\operatorname{sen}^2\theta_0}$$

$$= \underbrace{m\dot{r}^2 + mr^2\operatorname{sen}^2\theta_0\dot{\phi}^2}_{T} \underbrace{- \frac{1}{2}m\dot{r}^2 - \frac{1}{2}mr^2\operatorname{sen}^2\theta_0\dot{\phi}^2 + mgr\cos\theta_0}_{\neq V}$$

$E = T + V \overset{?}{=} H$

Acá el H no es la energía del sistema.

$$= m\left[\frac{p_r^2}{m^2}\right] + mr^2\operatorname{sen}^2\theta_0\left[\frac{p_\phi^2}{(mr^2\operatorname{sen}^2\theta_0)^2}\right] - \frac{1}{2}m\left[\frac{p_r^2}{m^2}\right] - \frac{1}{2}mr^2\operatorname{sen}^2\theta_0\left[\frac{p_\phi^2}{(mr^2\operatorname{sen}^2\theta_0)^2}\right] + mgr\cos\theta_0$$

Tenemos ...

$$H(r,\phi,P_r,P_\phi,t) = \frac{2}{2}\frac{P_r^2}{m} - \frac{1}{2}\frac{P_r^2}{m} + \frac{2}{2}\frac{P_\phi^2}{mr^2\operatorname{sen}^2\theta_0} - \frac{1}{2}\frac{P_\phi}{mr^2\operatorname{sen}^2\theta_0} + mgr\cos\theta_0$$

$$\vdots$$

$$H(r,\phi,P_r,P_\phi,t) = \frac{P_r^2}{2m} + \frac{P_\phi^2}{2mr^2\operatorname{sen}^2\theta_0} + mgr\cos\theta_0$$

Ecuaciones canónicas de Hamilton:

$$\left.\begin{aligned} \dot{P}_i &= -\frac{\partial H}{\partial q_i} \\ \dot{q}_i &= \frac{\partial H}{\partial P_i} \end{aligned}\right\}$$ Producen 2n EDO de primer orden

Tenemos

$$\hookrightarrow \dot{P}_r = -\frac{\partial H}{\partial r} \quad ; \quad \dot{r} = \frac{\partial H}{\partial P_r}$$

$$\dot{P}_\phi = -\frac{\partial H}{\partial \phi} \quad ; \quad \dot{\phi} = \frac{\partial H}{\partial P_\phi}$$

Tenemos

$$\dot{P}_r = \frac{P_\phi^2}{mr^3\operatorname{sen}^2\theta_0} - mg\cos\theta_0 \quad ; \quad \dot{r} = \frac{P_r}{m}$$

$\dot{P}_\phi = 0 \Rightarrow P_\phi = cte$ en el tiempo ; $\dot{\phi} = \dfrac{P_\phi}{mr^2\operatorname{sen}^2\theta_0}$

$\phi$ = coordenada cíclica

(si es cíclica en la Lagrangiana también debe de heredarse a la Hamiltoniana)

Tenemos así 4 ecuaciones ...

$$\dot{P}_r = \frac{P_\phi^2}{mr^3\operatorname{sen}^2\theta_0} - mg\cos\theta_0 \quad ; \quad \dot{r} = \frac{P_r}{m}$$

$$\dot{P}_\phi = 0 \quad ; \quad \dot{\phi} = \frac{P_\phi}{mr^2\operatorname{sen}^2\theta_0}$$